Calculate the message digest of a text using the SHA-1 algorithm in JAVA | Cryptography

HARDWARE AND SOFTWARE REQUIREMENT:

1. Intel based Desktop PC: - RAM of 512 MB

2. Notepad/Notepad ++ editor

3. Net beans / Eclipse

THEORY:

Ø A hash function is a special mathematical function that converts a numerical input value of arbitrary length into another compressed numerical value(fixed length). A hash performs one-way encryption.Values returned by a hash function are called message digest or simply hash values. hash codes or simply hashes.

Ø It is fundamental part of password security, in terms of database storage. A hash function responsible for changing a text password into a more complex set of characters by using more complex operations than a familiar mathematical operation such as multiplication.

Ø Using hashes and hexadecimal formats can help those who are storing passwords on a database to confuse hackers.

Ø The Secure Hash Algorithm 1 (SHA-1) is a cryptographic computer security algorithm.SHA-1 produces a **160-bit** hash value from the inputted data (data that requires encryption).

Ø SHA-1 is commonly used in cryptographic applications and environments where the requirements for data integrity is high. Some of the protocols that use SHA-1 include:

· Transport Layer Security (TLS)

· Secure Sockets Layer (SSL

· Pretty Good Privacy (PGP)

· Secure Shell (SSH)

· Secure/Multipurpose Internet Mail Extensions (S/MIME)

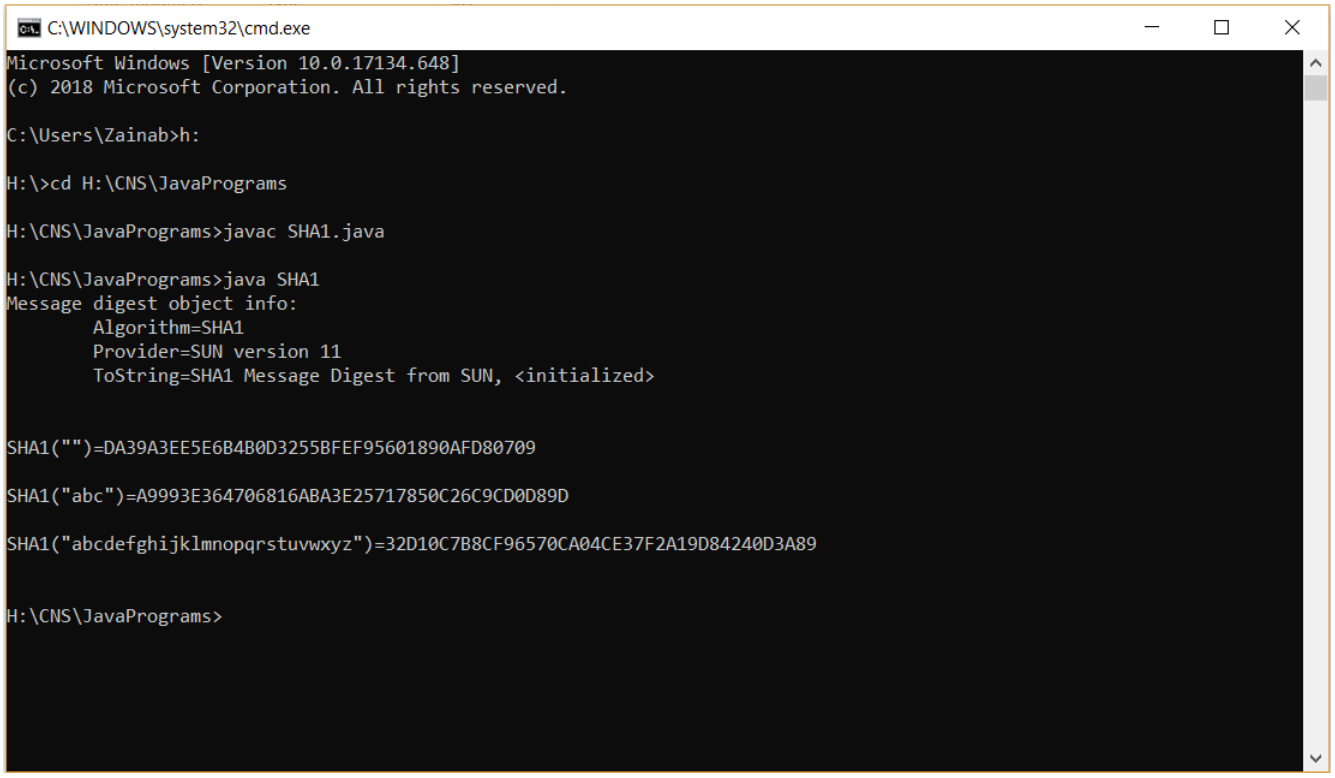
· Internet Protocol Security (IPSec)

SOURCE CODE:

```
import java.security.*;

public class SHA1 {

   public static void main(String[] a) {

      try {

         MessageDigest md = MessageDigest.getInstance("SHA1");

         System.out.println("Message digest object info:");

         System.out.println("         Algorithm=" + md.getAlgorithm());

         System.out.println("         Provider=" + md.getProvider());

         System.out.println("         ToString=" + md.toString());

         String input = "";

         md.update(input.getBytes());

         byte[] output = md.digest();

         System.out.println();

         System.out.println("SHA1(\"" + input + "\")=" + bytesToHex(output));

         input = "abc";

         md.update(input.getBytes());

         output = md.digest();

         System.out.println();

         System.out.println("SHA1(\"" + input + "\")=" + bytesToHex(output));

         input = "abcdefghijklmnopqrstuvwxyz";

         md.update(input.getBytes());

         output = md.digest();

         System.out.println();

         System.out.println("SHA1(\"" + input + "\")=" + bytesToHex(output));

         System.out.println();
```

```
        } catch (Exception e) {
            System.out.println("Exception:" + e);
        }
    }
    private static String bytesToHex(byte[] b) {
        char hexDigit[] = {'0', '1', '2', '3', '4', '5', '6', '7', '8', '9', 'A', 'B', 'C', 'D', 'E', 'F'};
        StringBuffer buf = new StringBuffer();
        for (int j = 0; j < b.length; j++) {
            buf.append(hexDigit[(b[j] >> 4) & 0x0f]);
            buf.append(hexDigit[b[j] & 0x0f]);
        }
        /*
        for loop in optimized way
        for (byte aB : b) {
            buf.append(hexDigit[(aB >> 4) & 0x0f]);
            buf.append(hexDigit[aB & 0x0f]);
        }
        */
        return buf.toString();
    }
}
```

OUTPUT:



VIVA QUESTIONS:

1. What is JAVA?
2. What is SHA-1 Algorithm?
3. What is java script?
4. What is Cryptography?
5. What is Encryption?
6. What is Decryption?